

Theta Gate — pre-flight analysis, 30 Aug 2026

Theta Gate team

30 August 2026

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Theta Gate — analysis and Sunday fixes, 30 Aug 2026

Status after this session (Sun 30 Aug, ~14:30 Riyadh / 07:30 ET). The audit below was produced by a 157-agent read-only workflow (11 finders → 3-lens adversarial verification → 4 judges → 2 designs → critic → synthesis), then the Sunday fix set was implemented by a second workflow, adversarially re-verified, and repaired. `pytest -q`: **96 → 136 passed**. The throwaway `paper` account was used for one submit/lookup/duplicate/cancel probe (order `tg-probe-20260830-s0`, canceled, account flat); the submission account was never written to.

What landed on `main` today

Commit	Change	Finding
<code>1e980f4</code>	<code>.memsearch/</code> untracked and ignored	F21
<code>5232566</code>	<code>LICENSE</code> (MIT) + <code>.agents/skills/LICENSE</code> (Apache-2.0) + <code>NOTICE</code> + README line	hackathon-compliance-2
<code>8d48c9e</code>	<code>ci.yml</code> : <code>pytest</code> on push/PR	tests-coverage-10
<code>0b21c85</code>	<code>docs/hackathon-rules-2026-08-30.md</code> archived (Gate 0)	—
<code>9227443</code>	Option chain: <code>--limit 1000</code> + <code>next_page_token</code> ; strike window <code>spot×0.90..1.02</code> at entry, legs ± 0.50 at exit	F1, exit-path-2
<code>5d4764f</code>	<code>alpaca._run</code> reads the CLI's <code>stderr</code> error body → <code>AlpacaCLIError</code> ; 404 lookup = miss; <code>_lookup_or_submit</code> skips dead orders (walk <code>r2..r6</code>), journals <code>submit_failed</code> ; force rung tags carry the date	entry-path-1, F2, F19, F20, broker-contract-2
<code>b6e5ffe</code>	Journal published on every tick path (exception, <code>not_paper_abort</code>)	entry-path-3, durability-runner-2
<code>f965522</code>	<code>vrp.realised_vol_lookback_days</code> is now read (no behaviour change at 20)	strategy-pnl-1 (mechanism only)
<code>a1d39fc</code> <code>ac8c63c</code> <code>6fa9170</code>	Cancel any live sibling entry/exit order before a ladder starts (re-verify found the id walk could submit beside a crashed tick's live <code>s1</code>)	regression caught in re-verify
<code>920e687</code>	<code>timeout-minutes: 25</code> ; <code>gate_check</code> dispatch runs <code>scripts/live_gate_check.py</code> , which now fails loudly	F4, F11
<code>f5ed41e</code>	Review hardening: trailing CLI text, id-less hits, page-cap overflow, git errors, zero-bid exit legs	re-verify lows/mediums

One deliberate divergence from the synthesized plan. Rank 3 below proposes a `window_already_attempted` guard (one fill attempt per window). What landed is the reviewer-endorsed id walk (`s0 → s0r2 ... r6`): each tick inside a window gets a fresh attempt at a fresh mid. The cost is one billed `brain.propose` per tick in the window ($\sim \$0.04$); the benefit is up to three fill attempts per window instead of one.

Refuted or downgraded by the verifiers, so not done: `requirements-dashboard.txt` + Python pin (F22 refuted — Streamlit Cloud runs 3.14); the `store.rebuild` race (F9 downgraded — Streamlit's per-key compute lock narrows it; `unlink(missing_ok=True)` stays a Monday item).

Decisions only PK can make

#	Decision	Numbers	Recommendation
1	VRP gate (<code>governance.json</code> → <code>vrp</code>). On Friday's close every SPY/QQQ candidate is vetoed: ATM IV 8/9-DTE SPY 9.9 / 10.1 % vs RV20 10.4 % ; QQQ 14.8 / 15.2 % vs RV20 18.2 % ; floor +2.0. RV20 still carries the 3–4 Aug rally days; they roll out Wed.	A: keep 20 / 2.0 → no entries Mon–Tue, first plausible Wed 10:30 (one window all week). B: lookback 10 / floor 2.0 → SPY +2.1..+2.3 (thin), QQQ +1.3..+1.7 (no). C: lookback 10 / floor 1.0 → both pass on Friday's marks. D: disable the gate → trades, but abandons the only claimed edge.	C : tenor-match the window to the 6–9 DTE horizon and re-base the floor with it. Two numbers in <code>governance.json</code> (<code>realised_vol_lookback_days: 10</code> , <code>min_vrp_points: 1.0</code>) + the <code>_comment</code> , pushed before Mon 10:25 ET . It is a strategy change on the “only edge” claim — public positioning — hence yours.
2	LICENSE copyright holder	<code>LICENSE</code> says “Theta Gate contributors (github.-com/Kalwaleed/theta-gate)”	Rename to the entity you submit under, or leave.
3	Commit-history disclosure	<code>96fd434</code> (alpaca/risk/spread/governance/tests, 1,049 lines) landed 28 Aug 07:20 EDT, 3h40m before kickoff; 60–73 % of those files still date from it	Say it in the write-up; judges check history. Wording is yours.
4	Public flip + PR triage	Repo PRIVATE; PRs #7 (store races), #8 (temporary cron probe — now stale, touches <code>agent.yml</code>), #9 (dashboard: HTML injection fix + thesis rows — a judge lever), #10 (ECC bundle again), #11 (deck; edits <code>.gitignore</code>), #12 (cover)	Merge #7 and #9 Monday after close (rebase onto today's <code>.gitignore /README</code>); close #8 and #10; hold #11/#12 for Thursday's numbers. Flip public Tue evening after <code>/security-review</code> .
5	Submission account	Created 28 Aug 07:54 EDT, 3h06m before kickoff; \$100k, zero orders	Keep. Re-creating changes the judged id and every GitHub secret.

Monday

- **Pre-market**: if C is chosen, push the two-number governance edit by 10:25 ET; dispatch `gate_check=true` at ~09:45 ET and read the gate tally; watch the 09:30 tick (the first scheduled run in the repo's history) and the 10:30 window.
- **After close**: the report's Monday block — phantom-open reconciliation (F3), `assert_paper` once per poll loop (F4), qty 1→2 with the partial-fill design X1 (F18), `available_underlyings` in the brain context (strategy-pnl-2), the stale-order sweep outside windows (schedule-time-1), the dashboard batch (F5/F9/F10 + the new event names), then the MCP reconciliation script (X2, new workflow + `uv` → plan review).

Theta Gate — pre-flight analysis, Sun 30 Aug 2026

HEAD 92b2f96 == origin/main, 96 tests green, submission account 7a013821-... ACTIVE with zero orders, working tree dirty only with .memsearch noise and two untracked scratch files. Open PRs: #7, #8, #9, #10, #11 (#2 is not open).

Verdict

As shipped, zero orders all week: every submit path dies at the first client_order_id lookup (CLI 0.0.13 writes the 404 to stderr; alpaca._run reads stdout only). After that fix, the VRP gate still vetoes both underlyings Mon/Tue on Friday's data; the first plausible entry is Wed 2 Sep 10:30 ET. Tonight's five code changes make trading possible; the VRP floor is PK's call before Mon 10:25 ET. The submission is not judgeable today: private repo, no LICENSE, no demo URL, no video, slides, cover or write-up. Fix the broker boundary tonight, flip public and deploy Tue evening, build media Wed–Thu. What wins is a visible proposer-vs-guard record and an honest write-up, not P&L (ceiling $\approx$ +\$260 gross).

Judge scorecard

Criterion	Theta Gate today	Best competitor (score)	The one move
P&L Performance	2	AlphaPilot AI / NewsFlow Trader / Vega / continual-learning agent / Options Sniper (5)	Rank 1 broker-boundary fix tonight, then PK's VRP floor call before Mon 10:25 ET $\rightarrow$ 3 (Wed-only entry) or 4-5 (Mon–Thu round trips at a relaxed floor)
Technology Implementation	3	Vega — autonomous long-gamma options agent (8)	Rank 1 + public repo (rank 28) + judge-visible MCP reconciliation (rank 22) $\rightarrow$ 6
Creativity & Originality	5	Tissue Regeneration Navigator / Odysseus / Vega (7)	Proposer-vs-Guard ledger on the dashboard (rank 21) and "MCP is the auditor, not the trader" framing $\rightarrow$ 6
Presentation & Execution	2	BABIL / NewsFlow Trader / Vega (7)	Public repo + hosted Streamlit URL + PR #9 thesis rows (rank 28) $\rightarrow$ 5; video/deck/write-up (ranks 31–33) decide the rest

Competitor scores come from lablab blurbs only; none shows P&L evidence.

Confirmed findings

48 verifier records (34 at 3/3, 4 at 2/3, 10 at 1/1) collapse to the rows below. "Where" is corrected where the critic found a bad citation.

id (votes)	severity	where	claim	fix	crit- ter- ion
entry-path-1 (3/3)	critical	alpaca.py:36-45, 170-178; loop.py:288	CLI 0.0.13 writes every error to stderr, exit 1, empty stdout; <code>_run</code> json-loads stdout only, so the <code>client_order_id</code> miss before every submit raises <code>RuntimeError</code> ; no order can ever be placed	<code>_run</code> raises typed <code>AlpacaCLIError</code> from stderr JSON; 404 → None in <code>get_order_by_client_id</code> ; delete the <code>code == 0</code> filter (rank 1)	pnl
tests-cover-age-1 (3/3)	critical	alpaca.py:36-45; test_loop.py:67,76,81	Same root; the suite mocks the lookup to return None, a value the real function cannot produce	Tests at <code>subprocess.run</code> with the observed <code>CompletedProcess</code> shapes (rank 1)	pnl
F2 (2/3 crit, 3/3 high, 3/3 med)	critical	loop.py:291-298; alpaca.py:36-45; test_loop.py:75-84	A rejected submit never reaches the <code>dry_run</code> label — it raises in <code>_run</code> with an empty message, tick dies unpublished; the <code>dry_run</code> test pins an unreachable contract	Return (payload, 'rejected') for definitive 4xx, re-raise 5xx/timeout, classify <code>dry_run</code> by the flag (rank 1)	pnl
F1 (3/3 x3)	critical	alpaca.py:131; market.py:267; loop.py:376	<code>--limit 100</code> , no token, no strike window: one expiry only, top strike SPY 773 / QQQ 731 (0.47% above spot → <code>atm_iv</code> None → <code>vrp veto</code>), exit fetch shares the cap and silently skips every exit incl. Thu rungs	<code>--limit 1000</code> + token loop in <code>option_chain</code> , strike windows at both callers with the upper bound above spot (rank 2)	pnl
strategy-pnl-1 (3/3)	critical	risk.py:207-223; governance.json:77-83	On Friday's data VRP is $-0.3/-0.5$ (SPY) and $-3.1/-3.4$ (QQQ) vs floor 2.0; every candidate vetoed Mon/Tue; Wed 10:30 first plausible pass	PK decision on the floor (rank 7); RV10 is a 3-file code change, not a one-liner	pnl
entry-path-2 (3/3)	high	governance.json:77-83 (not 83-88)	Same gate; <code>min_vrp_points=0</code> still vetoes SPY; <code>flag=false</code> skips the missing-IV guard	Use a negative floor if relaxing; keep the flag (rank 7)	pnl
F3 (3/3 x3)	high	loop.py:533-538, 548-551, 604-618	Force-close rungs never poll; every inter-tick fill is journaled as <code>exit_position_already_flat</code> , never <code>exit_filled</code> ; position stays "open" with pnl None at submission	<code>list_orders(closed, limit 500)</code> filter + summed <code>filled_qty</code> → <code>_journal_exit_fill</code> ; no synthetic record (rank 17)	pre- sen- tation
F4 (3/3 high, 2/3 high, 3/3 low)	high	alpaca.py:181-183, 235-245; loop.py:301-307	Every poll = doctor + account get + order get + 2 s; worst tick ~10.6–11.8 min vs timeout 10; kill loses the publish	timeout 25 (rank 4); assert once per poll loop + monotonic deadline (rank 18)	pnl
F18 (3/3)	high	loop.py:696-723	At qty 2 a canceled partial s0 is journaled nowhere, symmetry is	X1 partial-fill handling (rank 16)	pnl

id (votes)	severity	where	claim	fix	crit- ter- ion
			symbol-only, s1 resubmits full qty → 3 lots, no HALT		
entry-path-3 (3/3)	high	loop.py:925-936, 946-948, 1043	<code>_git_publish</code> only on the happy path; exception and not- paper_abort ticks die with the runner; a hang after submit leaves a live day order	Single publish in <code>_run_tick_guarded</code> after try/except (rank 5)	tech
durability- runner-2 (3/3)	high	loop.py:931-936, 947-948, 1043	Same; <code>_consecutive_exceptions</code> is identically 0 on the runner; the 3-strike gate is an entry veto, not a HALT	Same (rank 5)	tech
exit-path-2 (3/3)	high	loop.py:372-381, 634-638	A leg missing from the capped fetch → <code>exit_quote_unavailable</code> (no level, not whitelisted) → returns before <code>exit_signal</code> , all four Thu rungs and the 15:45 alert	Strike-bounded exit fetch (rank 2); level + whitelist (rank 21)	com- pli- ance
broker-con- tract-2 (3/3)	high	loop.py:310-323; alpaca.py:170-178	Cancel-after-fill 422 goes to <code>stderr</code> → <code>_run</code> raises before the poll; the raced-fill branches are unreachable; fill unjournalled → HALT next tick	Catch status 422 only in <code>_cancel_and_confirm</code> (rank 1)	pnl
schedule- time-1 (3/3)	high	loop.py:1023-1026 (step 9), 284-298, 571	A tg-e-* order left working by a dead tick is never looked up or canceled; rests to 16:00; a late fill HALTs and is unmanaged	Tick-start sweep through <code>_cancel_and_confirm</code> with fill reconstruction (rank 19)	pnl
schedule- time-2 (3/3)	high	loop.py:984-993, 995-1000	An untracked broker position is HALTed but never exit-managed or flattened Thu	Reconstruct entry_filled from the closed tg-e- order (rank 27)	pnl
tests-cover- age-4 (3/3)	high	loop.py:430-744	12 of 35 loop.py functions run under tests; none of the eight order-path functions	X1 ships with 21 tests (rank 16); CI pytest (rank 12)	pnl
strategy-pnl- 2 (3/3) / en- try-path-4 (1/1)	high	loop.py:769-803; brain.py:113-126	Brain never sees open/filled underlyings; pipeline tries only the pick; a repeat SPY after a fill wastes every later window and bills propose	available_underlyings in context + deterministic no_trade backstop (rank 20)	pnl
docs-vs- code-2 (3/3)	high	docs/diagrams/architec- ture.html (body, not line 1)	Two LLM roles on “read-only MCP”, a Critic, 14 gates, 15-min cron; same in flowchart/sequence, README:5, PLAN:79/102	Redraw all three via diagram-design (rank 24)	pre- sen- tation
hackathon- compliance- 1 (3/3)	high	<code>gh repo view</code> (PRIVATE), not README:1	Public repo is mandatory; flip needs .memsearch untrack,	Ranks 8, 9, 28	com- pli- ance

id (votes)	severity	where	claim	fix	crit- ter- ion
			LICENSE, log review; history residue is permanent		
hackathon-compliance-2 (3/3)	high	<code>ls LICENSE*</code> empty; .agents/skills (3,439 Apache-2.0 lines)	No LICENSE; ToS requires MIT; vendored skills lack the Apache text	MIT + Apache-2.0 + attribution (rank 9)	compliance
hackathon-compliance-6 (3/3)	high	app.py (undeployed)	Application URL is required; nothing technical blocks deploying main as-is	Deploy from main after the flip (rank 28)	compliance
hackathon-compliance-7 (3/3)	high	docs/HANDOFF.md:22; PR #11	No MP4, no PDF on main, no 16:9 cover, no write-up; deck contradicts the fixed decisions	Ranks 25, 29, 31, 33	presentation
F5 (3/3 x2)	medium	store.py:449-453; app.py:387-404	Whitelist drops 5 of 9 critical events + the only warning; HALT reason never rendered; [:60] buries criticals	level-based WHERE + alerts panel + visible reason (rank 21)	presentation
F19 (3/3, 2/3)	medium	loop.py:284-298, 700-703, 735-736	Ticks 2–3 of a window re-adopt the canceled s0/s1, re-run the pipeline and bill propose; re-cancel is idempotent (rc=0 '{}' observed)	window_already_attempted guard + terminal-status cancel guard (rank 3)	pnl
tests-coverage-2 (2/3)	medium	test_loop.py:65-72	Test pins adoption of ANY hit incl. canceled	Same (rank 3)	pnl
F21 (3/3 x2)	medium	.memsearch/.index-state.json; .memsearch/memory/2026-08-26.md	Tracked notes carry 14 transcript paths, username, ~/.zshrc edits, credential-store path, the negative-EV note; two more note files one <code>git add -A</code> away	<code>git rm -r --cached</code> + <code>.memsearch/</code> ignore, accept history (rank 8)	compliance
docs-vs-code-1 (3/3)	medium	docs/diagrams/sequence.html (body)	Depicts a BEAR CALL fill and a 0.45 credit/width veto, 14 gates, two proposals per tick — all impossible	Redraw (rank 24)	presentation
F8 (1/1)	medium	loop.py:955	open_orders never reaches gate_concurrent; docstring lies	Sweep (rank 19), not a fold into open_positions	none
F11 (1/1)	medium	agent.yml:104-128	brain.propose, Cboe fetch and chain fetch have never run on the runner; first run is Mon 10:30	gate_check smoke job (rank 6)	pnl
exit-path-3 (1/1)	medium	loop.py:455, 496-511	Exit s1 resubmits the pre-partial qty; partial debit never journaled	X1 exit_partial_fill (rank 16)	pnl
exit-path-4 (1/1)	medium	loop.py:548-551	15:45 rung is a journal line only: green run, no HALT, filtered from the dashboard	Summary ok=False + whitelist by level (ranks 17, 21)	compliance

id (votes)	severity	where	claim	fix	crit- teri- on
broker-con- tract-3 (1/1)	medi- um	loop.py:925-936, 141-151	Exception path never publishes; the breaker is blind; error text is empty	Rank 5 + stderr[:500] in rank 1	pre- sen- tation
F6 (1/1)	low	loop.py:71	A 15-min window is missed only by a ≥15-min scheduler hole	Optional 25 min with a .get default (rank 23); runbook fallback (rank 11)	pnl
F7 (1/1)	low	loop.py:1039-1057	tick_completed ok=True before publish; comment is false; moving it after publish would lose every heartbeat	Fix the comment only (rank 5)	none
F10 (1/1)	low	app.py:358 (inside load(), not the governance render)	Unguarded governance read blanks the page	empty() + st.stop(); pytest before pushing governance edits (rank 21)	pre- sen- tation
F20 (1/1)	low	loop.py:546	Dateless rung tag re-adopts Thursday's terminal order on Friday	Date-stamped tag with the correct <code>now</code> variable (rank 3)	none

ENTRY-PATH-1 / TESTS-COVERAGE-1 (CRITICAL)

Observed live on the submission profile: `alpaca order get-by-client-id --client-order-id tg-e-x` → rc 1, stdout 0 bytes, stderr `{"code":40410000,"status":404,...}`; calling `alpaca.get_order_by_client_id('tg-e-x')` raised `RuntimeError: ... returned non-JSON:` . CLI source at v0.0.13 (root.go:50-83 printJSONError → os.Stderr; client.go:270) shows one error path for every command. `_lookup_or_submit` (loop.py:288) runs that lookup before `submit_mleg` and before the `dry_run` branch, so all five submit sites die. First bite: Mon 31 Aug 10:30 ET on the first gate-passing candidate; Thu 3 Sep 14:30 ET rung 1 identically.

F2 (CRITICAL → COROLLARY)

The `dry_run` misclassification at loop.py:297 is unreachable because a rejection raises one call earlier; the residual after rank 1 is that an uncaught submit rejection still aborts the tick and, on Thu, a persistent rejection on position 1's close starves position 2's rung (the exception exits the exit loop at loop.py:966-971). Fix objections agreed: raise a typed error, return (payload,'rejected') only for definitive 4xx, never continue the ladder on 5xx/timeout (a landed s0 plus an s1 is two live spreads).

F1 (CRITICAL)

Live: SPY puts 6–9 DTE at `--limit 100` → 100 snapshots, one expiry (260908), strikes 500–773, token set; `--limit 1000` → 330 across 260908/260909, token empty; 1000 is the API max (400 above it); `--page-token` and decimal `--strike-price-gte/lte` verified. Scratch: spot 773.03 → compute_atm_iv None → vrp veto; the proposed entry window spot×0.85..spot reproduces the same None (fixed by an upper bound above spot). Exit path: 260911 page 1 ends at 751 while its band shorts are 753–759 — a Wed 9/11 entry would be un-exitable at today's prices once the entry cap is lifted alone; both callers must change together.

STRATEGY-PNL-1 / ENTRY-PATH-2 (CRITICAL / HIGH)

Production path re-run on Friday's bars and chains (spread.parse_chain → market.compute_rv20 → rank_candidates → compute_atm_iv → risk.check_all) with live Cboe VIX 14.43 / VIX9D 11.22 / VIX3M 17.48: SPY 759/754 credit 0.605, ATM IV 0.0993 vs RV20 0.1040 → "VRP -0.5 points below floor 2.0"; QQQ 702/697 → -3.4. Every other gate passes. Roll-forward with flat days: RV20 → SPY 0.067 / QQQ 0.124 by Wed, so Wed 10:30 likely passes for SPY at Friday's IV if $IMonI,ITuel \leq 0.85\%$.

F3 (HIGH)

Scratch tests through `_evaluate_and_exit_position` with both legs gone: only `exit_position_already_flat` is journaled; `_open_positions` (loop.py:114-123) closes on `exit_filled` alone; `store._build_positions` mirrors it (status open, realised_pnl None). Paper mleg fills observed 73–96 ms after submitted_at on 26 Aug, so every rung fill is inter-tick by construction. Objections rejected the cid-guessing lookup (misses raise) and the synthetic close (fabricated P&L, transient empty read → permanent HALT).

F4 (HIGH)

Counted from the real functions with a fake broker: an unfilled exit ladder is 216–246 subprocesses + 120–140 s sleep; entry ladder 126–156 + 60–80 s. Runner latency from three Saturday runs: 0.24–0.52 s per spawn. The two-exits-plus-entry combination is blocked by `gate_concurrent`; the reachable worst case is 7–9 min plus ~30 s setup, straddling 10 min at the slow bound. Timeout 25 costs nothing; the poll refactor cuts 2/3 of spawns.

F18 / EXIT-PATH-3 (HIGH / MEDIUM)

Scratch on current code at qty 2: s0 canceled with filled_qty '1' → nothing journaled, `_check_leg_symmetry` passes (symbol-set test), s1 submits qty 2 → broker 3 lots, journal qty 2, max loss \$1,323 vs the \$880 gated, no HALT (untracked check keys on symbols). Exit variant: s1 resubmits qty 2 against 1 held pair. Unreachable at qty 1; live from Tue 1 Sep 10:30 ET.

ENTRY-PATH-3 / DURABILITY-RUNNER-2 / BROKER-CONTRACT-3 (HIGH / HIGH / MEDIUM)

`_git_publish` has one call site (loop.py:1043). Scratch: body raising → `git` never spawned; `not_paper_abort` return → never spawned. `_consecutive_exceptions` counts only tick_completed ok=False rows on main, so it is identically 0 for runner failures; risk.py:329 is an entry veto inside `check_all`, not a HALT. A `finally` would double-publish; the single-site move covers all paths.

EXIT-PATH-2 (HIGH)

Live: SPY 260908 puts at `--limit 100` = strikes 500–773 (band shorts 757–761 inside, 12 strikes of headroom); QQQ 490–731 (27 strikes). A missing leg never surfaces as a CLI error (empty expiry returns rc 0, snapshots {}); the only trace is the un-whitelisted `exit_quote_unavailable` line while ticks stay green.

BROKER-CONTRACT-2 (HIGH)

`order cancel --help`: "no longer cancelable ... rejected with status 422"; the CLI's single error printer routes it to stderr (observed for 404 on the submission profile); loop.py:321 has no handler. Race window ≈ 1–3.4 s per cancel (assert_paper spawns doctor + account get before the DELETE). A cancel of an already-canceled order returned rc 0 {} (observed 2026-08-30 08:26 UTC) — only the filled case raises.

SCHEDULE-TIME-1 / SCHEDULE-TIME-2 / F8 (HIGH / HIGH / MEDIUM)

Reproduced end-to-end through `run_tick` with the broker mocked (7 scratch tests): `open_orders` (loop.py:955) feeds only the force-close stale loop; a tg-e-* order left by a dead tick rests as a day order; a fill outside the window is untracked → HALT (loop.py:984-993) and the exit loop (loop.py:995) never iterates it, so Thu 14:30 skips it. HALT is re-derived from broker state, so editing HALT.json does not clear it.

TESTS-COVERAGE-4 (HIGH)

Measured with a settrace/setprofile hook: 12 of 35 loop.py functions execute under the suite; none of `_attempt_entry`, `_attempt_exit`, `_attempt_force_close`, `_evaluate_and_exit_position`, `_journal_entry_fill`, `_journal_exit_fill`, `_wait_for_terminal`, `spread.closing_mleg_body`. No pytest step exists in CI.

STRATEGY-PNL-2 / ENTRY-PATH-4 (HIGH / MEDIUM)

Scratch through the real `_attempt_entry_pipeline` and `_run_tick_body`: `brain_context` carries `spot/rv20/move/vix` only; after a SPY entry `_filled`, a SPY proposal ends at `gate_concurrent` (“already at max positions for SPY”) with `rank_candidates` never called for QQQ; up to 5 billed ticks on fill day. The model repeating SPY is a premise, not an observation.

DOCS-VS-CODE-2 / DOCS-VS-CODE-1 (HIGH / MEDIUM)

Tag-stripped text of the three diagrams: ‘Critic’ x3, ‘14 gates’, ‘every 15’, ‘read-only MCP’ x2, ‘BEAR CALL’, ‘0.45’, ‘Fourteen’ x5. Code: `brain.py:235-244` `tools=[]` `mcp_servers={}` `strict_mcp_config=True` `max_turns=1`; `risk.py:349-374` = 21 gates; `agent.yml:20-22` every 5 min; `resolve_direction(‘bearish’)` → None (7 scratch tests). `HANDOFF.md:77` wrongly records the diagram as corrected.

HACKATHON-COMPLIANCE-1 / -2 / -6 / -7 (HIGH)

Re-verified by the critic: `gh repo view` → PRIVATE, `licenseInfo` null, `topics` null; no LICENSE*; 0 deployments; only a 13-page Beamer deck in PR #11 with 6 PLACEHOLDER stats. Live rule book (fetched 30 Aug): public repo mandatory, MP4 + PDF mandatory, 16:9 cover, video under 3 min scores 2/5. Streamlit Cloud deploys main as-is (3.12 default, 24/24 dashboard tests).

Refuted findings

id	title	why
F2	A rejected/errored submission is classified as a dry-run	CLI errors go to <code>stderr</code> with empty <code>stdout</code> (6 read-only probes + <code>source root.go/client.go</code>); <code>_run</code> raises before <code>loop.py:297</code> ; the same shape breaks the lookup one call earlier — the real critical (<code>entry-path-1</code>)
F4	One unfilled exit ladder ≈ 6.5 min; two simultaneous stop exits cannot finish in 10 min	Mechanics confirmed (216–246 spawns + 120–140 s per ladder) but the 3.6 s/call is Gulf-to-US latency; runner spawns are 0.24–0.52 s (three real runs) → 6.5–8.7 min end to end; <code>force-close</code> does not poll
F4	<code>assert_paper</code> triples per-call cost: ~3.3 s/call, a 60-attempt poll is 5.3 min	No caller passes 60 attempts (<code>loop.py:306</code> caps 30/15); runner latency ≤ 1.5 s per wrapped call; the compound tick is blocked by <code>gate_concurrent</code> ; realistic worst case 5–6.5 min
tests-cover-age-3	<code>order_cancel</code> returns 204 with no body, <code>_run</code> raises on every successful cancel	<code>voidResponse</code> (<code>factory.go:115-123</code>) prints <code>{}</code> rc 0; observed live twice (29 Aug throwaway, 30 Aug submission profile, idempotent on a canceled id); the unpinned branch is the 422 after a fill (<code>broker-contract-2</code>)
F9	<code>store.rebuild</code> TOCTOU breaks the demo under concurrent load	Reproducible only on bare <code>store.connect()</code> (236 errors/30 rounds); <code>app.load</code> sits behind <code>st.cache_data</code> whose per-key compute lock serialises cold loads (8 threads → 1 rebuild, 0 errors); residual sub-second cross-mtime window self-heals; PR #7 is hardening
F22	Streamlit Cloud: Python 3.14 and the 103 MB <code>claude-agent-sdk</code> block deployment	Cloud defaults to 3.12; <code>requirements.txt</code> resolves on manylinux 3.12/3.13 (58 packages); <code>test_app</code> 24/24 on 3.12; <code>app.py</code> never imports the sdk; do NOT split <code>requirements.txt</code> (<code>agent.yml:56</code> installs it, <code>brain.py</code> imports the sdk at module level)

Unverified

Low severity passed through, or medium beyond the per-dimension cap of 12. None was re-derived.

id	severi- ty	where	title
F12	low	loop.py:268-276	_map_account_state defaults
F13	low	risk.py:312-316	gate_daily_drawdown fails open on missing session_start_equity
F14	low	risk.py:377-392	gate_deadline runs last; Wed afternoon ticks still fetch chains and bill a propose
F15	low	spread.py:245-251	Dead code and unread governance keys
F16	low	README.md:5	README says 'read-only tools'; brain.py has no tools (fixed in rank 9)
F17	low	loop.py:353-365	filled_avg_price sign convention verified; pin with a test (rank 1)
entry-path-5	low	risk.py:199-203	Quote age measured against tick-start now
exit-path-5	low	loop.py:995-1000	One position's exit exception aborts the whole tick
exit-path-6	low	loop.py:559-569	market_mleg rung caps at width-0.01, below natural for deep ITM
broker-contract-5	low	alpaca.py:36-40, 67-75	Nested timeouts inverted; TimeoutExpired uncaught
schedule-time-6	low	loop.py:634-648	First tick evaluates stop/TP on 09:30 quotes with no sanity check
durability-runner-5	low	agent.yml:30-32, 41	Single pending slot: a long tick cancels the next (grey, not red)
durability-runner-8	low	brain.py:226-265	brain.propose verified locally in a runner-shaped env (9.5 s cold)
durability-runner-9	low	agent.yml:97-107	Runner CLI ignores the 'submission' profile; paper rests on LIVE_TRADE=false + assert_paper
read-model-dash-board-5	low	store.py:480-482	'Sessions of six' counts weekend rehearsals
F9	low	store.py:216-221	TOCTOU real in isolation; cache lock narrows exposure
F22	low	requirements.txt:12	Cloud Python default is not a blocker; only wheel weight
read-model-dash-board-6	low	docs/PLAN.md:332	'Actions pings each tick' mitigation does not exist
read-model-dash-board-7	low	app.py:373	page_icon 'Θ' is not an emoji
tests-coverage-11	low	test_agent.py:19-68	GOV fixture is a hand copy (fixed in rank 16)
tests-coverage-12	low	test_store.py:419-427	Nothing asserts CRITICAL events survive the whitelist (rank 21)
F17	low	loop.py:353-365, 662	Sign convention pinned end to end in scratch, absent from the suite

id	severity	where	title
tests-coverage-13	low	alpaca.py:63-84	assert_paper fail-closed branches untested
docs-vs-code-12	low	docs/PLAN.md:241	Small-fact drift: 14 assertions, 44/23 tests, 4–9 DTE, ‘no chase’
docs-vs-code-13	low	README.md:5	Headline claims: read-only tools, proposer and critic, asserted at import
docs-vs-code-14	low	docs/STRATEGY.md:3	SUPERSEDED banner admits two stale items; ≥12 more contradict governance
docs-vs-code-15	low	social/README.md:19	Post 01 status contradicts itself across three files
docs-vs-code-16	low	app.py:509	‘rendered verbatim’ while six governance keys are never read
docs-vs-code-17	low	risk.py:110	Tier-1 blackout docstring wider than market.py implements
docs-vs-code-18	low	docs/HANDOFF.md:59	‘No blockers remain’, sign ‘still unverified’ — overtaken
docs-vs-code-19	low	app.py:419	Sessions counts every journal date
docs-vs-code-20	low	docs/PLAN.md:216	Claims ranking across every expiry; —limit 100 sees one
security-public-repo-3	low	app.py:493-499, 455-456, 313	Unescaped journal strings in unsafe_allow_html; PR #9 fixes
security-public-repo-4	low	.mcp.json:3-5	alpaca-mcp-server unpinned via uvx (pinned in rank 22)
security-public-repo-5	low	docs/HANDOFF.md:24-104; docs/research	Internal ops narrative goes public with the flip
security-public-repo-6	low	brain.py:113-126, 152-194	brain.py trust boundary has zero tests
hackathon-compliance-9	low	docs/HANDOFF.md:84	Manual remediation on the judged account needs a disclosure line
hackathon-compliance-11	low	docs/hackathon-rules-2026-08-30.md:311	Prize/platform terms bind PK personally (ID, W-8BEN, publicity rights)
hackathon-compliance-12	low	alpaca.py:36	Hard gates that PASS today with evidence
strategy-pnl-4	low	governance.json	P&L ceiling ≈ +\$260 gross at qty 2; realistic –\$600... +\$250
strategy-pnl-5	low	governance.json	credit_quality expected ratio 0.8×δ overstates the observed 0.59–0.67×δ
strategy-pnl-6	low	market.py:133-137	realised_vol_lookback_days is rendered but never read
broker-contract-4	medium	loop.py:301-307, 66-67	attempts = wait//2 assumes 2 s per attempt (rank 18 deadline)
schedule-time-3	medium	loop.py:925-937, 1043	3-strike halt and journal_publish_failed dead on CI (rank 5)

id	severity	where	title
schedule-time-4	medium	agent.yml:20-28	Schedule trigger has never fired; first at Mon 09:30 ET
schedule-time-5	medium	agent.yml:20-22	No end date on the cron (rank 34)
durability-runner-3	medium	loop.py:885-900	First <code>git pull --rebase</code> always fails on the dirty journal; one retry is the only sync
durability-runner-4	medium	loop.py:905-906, 1053	git timeout → committed=False → ok=True, journal silently lost (rank 5 treats error as failed)
durability-runner-6	medium	agent.yml:104-128	A red run is only an exit code; no notification
durability-runner-7	medium	market.py:35-44, 69-96	Cboe CSVs reachable from this host, never from a GitHub IP (rank 6)
read-model-dashboard-1	medium	store.py:449-453	Decision log flooded by outside_entry_window rows (rank 21)
read-model-dashboard-2	medium	store.py:139-146	'Why no trade' does not aggregate by gate
read-model-dashboard-3	medium	store.py:293-298	Realised P&L multiplies by entry qty (rank 16)
read-model-dashboard-4	medium	store.py:449-453	'proposal' events excluded; thesis lookup dead (rank 21, PR #9)
tests-coverage-5	medium	market.py:35-281	market.py has zero tests
tests-coverage-6	medium	brain.py:152-265	brain.py has zero tests
tests-coverage-7	medium	risk.py	Eight of 21 gates have no veto assertion
tests-coverage-8	medium	scripts/live_gate_check.py:72-95	Script does not mirror loop.py (rank 6 fixes the exit paths)
tests-coverage-9	medium	test_loop.py:87-94	Test cements 15-min window
tests-coverage-10	medium	agent.yml:37-128	pytest never runs in CI (rank 12)
docs-vs-code-3	medium	docs/diagrams/flowchart.html	'Fourteen gates'; regime group missing (rank 24)
docs-vs-code-4	medium	docs/PLAN.md:186	PLAN says cumulative drawdown writes HALT and closes positions; code returns a veto string
docs-vs-code-5	medium	governance.json:43	Rung 3 'market_mleg' is a limit order

id	severity	where	title
docs-vs-code-6	medium	docs/PLAN.md:189	Partial-fill unwind paragraph contradicts the code (rank 16 rewrites)
docs-vs-code-7	medium	docs/PLAN.md:82	Orphan equity 'flattened' — code waits for a human
docs-vs-code-8	medium	docs/PLAN.md:337	Rollback says <code>touch HALT</code> ; loop reads data/HALT.json
docs-vs-code-10	medium	governance.json:19	Seven 'exactly one contract' statements go false Monday (rank 16)
docs-vs-code-11	medium	docs/PLAN.md:251	'Artifact contract' names five non-existent files
docs-vs-code-21	medium	README.md:5	Built-but-unsurfaced strengths the write-up should lead with
security-public-repo-1	medium	git metadata	Personal Gmail on 14 of 47 commits (accepted in rank 8)
security-public-repo-2	medium	agent.yml:34-35	Third-party GitHub App holds push to unprotected main (rank 13)
hackathon-compliance-3	medium	social/README.md:19	Post 01 predates the window; 0 of 5 eligible posts (rank 25)
hackathon-compliance-4	medium	risk.py	60–73% of core code predates kickoff by 3h40m; disclosure is PK's call
hackathon-compliance-5	medium	docs/HANDOFF.md:30	Account created 3h06m before kickoff; history shows \$100k a day earlier (rank 13)
hackathon-compliance-8	medium	docs/HANDOFF.md:110	Deck/HANDOFF 'exactly one contract' (rank 16)
hackathon-compliance-10	medium	agent.yml:20	PR #8's Sunday cron probe is stale (rank 10 closes it)
strategy-pnl-3	medium	loop.py:674-744	If VRP first opens Wed, the week rides on one tick

Day plan

Skeleton corrections, stated explicitly: (a) the “_lookup_or_submit root cause” is in `alpaca._run`'s stdout-only parse, not in order adoption — the `r2..r6` suffix walk and the “submit_failed on a no-id response” are replaced by rank 1 and rank 3; (b) the chain fix's entry window must top out above spot (rank 2); (c) rank 5 (publish on every path) is added to Sunday; (d) `requirements-dashboard.txt` + Python pin is dropped (F22 refuted; splitting breaks Monday's tick); (e) `assert_paper` is not memoized per tick — assert once inside the poll loop (rank 18); (f) the phantom-open fix uses `list_orders`, not `cid_guessing`, and writes no synthetic close (rank 17); (g) PR #2 is not open — nothing to close.

SUNDAY

rank	action	owner	files	test	roll-back	PK decision
1	<code>_run</code> stderr → typed AlpacaCLIError; 404 → None; 422 cancel → poll; rejected → (payload,'rejected'); dry_run by flag; stderr[:500] in errors	Claude	alpaca.py, loop.py, test_agent.py, test_loop.py	Observed CompletedProcess shapes at subprocess.run: miss → None → 'submitted'; 422 submit → 'rejected', no s1; 422 cancel → raced fill journaled; 500 → raise; { } and " → { }; live get_order_by_client_id('tg- e-x') → None	git re- vert; back to fail- closed no-or- der state	no
2	option_chain --limit 1000 + token loop; en- try window spotx0.90..1.02; exit window long-0.5..short+0.5; docs/docstring fix	Claude	alpaca.py, market.py, loop.py, tests, PLAN/HANDOFF	181-row and 3-page fixtures: both expiries ranked, atm_iv at 773.5 not None, 759/754 and 9/11 753/748 exits resolve, kwargs reach argv	git revert	no
3	window_already_at- tempted guard; no can- cel on terminal status; date-stamped force tag with now.astimezone(ET)	Claude	loop.py, test_loop.py	10:35 tick with prior intent → no propose, no writes, ok=True; can- celed adopt → no cancel; Fri cid ≠ Thu cid	git revert	no
4	agent.yml timeout-min- utes 25	Claude	agent.yml	rehearsal dispatch shows the value	revert line	no
5	Single _git_publish in _run_tick_guard- ed after try/except; git error → publish_failed; fix comment	Claude	loop.py, test_loop.py	raising / NotPaperError / happy body each publish once; error dict → ok=False	git revert	no
6	gate_check dispatch job + live_gate_check.py ex- its (any MarketDataError → 1, None direction → 0, print exception, claude —version, Cboe curls, contracts > 100); then a dry_run rehearsal	Claude	agent.yml, scripts/live_- gate_check.py	green gate_check log with counts > 100 and the per-candidate Counter; dry_run rehearsal pushes the journal	re- move the job	no
7	VRP floor decision: keep 2.0 / negative floor / RV10 (defer); on main by 10:25 ET Mon	PK	governance.json	gate_check re-run Mon 09:45 ET	one- line revert	yes — strate- gy edge, public posi- tioning

rank	action	owner	files	test	roll-back	PK decision
8	<code>git rm -r --cached .memsearch</code> , ignore <code>.memsearch/</code> under Local scratch, ignore <code>firecrawl.err/hackathon_page.md</code> , commit rules archive, drop stash	PK	<code>.gitignore</code> , <code>.memsearch/*</code> , <code>docs/hackathon-rules-2026-08-30.md</code>	<code>git add -A -n</code> lists none of them	git revert (files stay on disk)	yes — history residue accepted
9	LICENSE (MIT verbatim) + <code>.agents/skills/LI-CENSE</code> (Apache-2.0) + README attribution + README:5 fix	PK	LICENSE, <code>.agents/skills/LI-CENSE</code> , README.md	licenseInfo MIT after the flip; no 'read-only tools'	delete files	yes — copyright holder
10	PR triage: merge #7; close #8; hold #10; #11 draft (freeze Tue); #9 Monday	PK	<code>store.py</code> , <code>test_store.py</code>	pytest green; #8 closed; 0 schedule runs	revert merge	no
11	<code>docs/RUNBOOK.md</code> (watch, manual dispatch <code>dry_run</code> unticked, HALT clear, naked-leg close, disclosure)	Claude	<code>docs/RUNBOOK.md</code>	every command copy-pasteable, none bulk	n/a	no
12	<code>ci.yml</code> running pytest on push/PR	Claude	<code>.github/workflows/ci.yml</code>	first push goes green	delete	no
13	PK checks: Anthropic key health, keep/re-create submission account, ecc-tools App access, keep Saturday journal lines	PK	—	account get ACTIVE; one <code>-p</code> call returns <code>claude</code>	n/a	yes — compliance, irreversible

MONDAY PRE-MARKET

rank	action	owner	files	test	rollback	PK decision
14	Throwaway 'paper' profile write rehearsal: submit (--dry-run and real), duplicate cid, cancel working (<code>{}</code>), after 09:30 fill → position list shape → cancel-after-fill 422 → close; fixtures into rank 1's tests	PK + Claude	test_agent.py, test_loop.py	fixtures replay exact rc/stdout/stderr; entry_filled credit == -filled_avg_price	close the throwaway position	yes — broker writes (paper)
15	Push VRP edit by 10:25 ET if chosen; confirm the first scheduled run 09:30; watch both windows; manual dispatch at 10:33 if silent; read red runs	team	—	schedule run exists by 09:35; window ticks show proposal + no_trade/entry_*, not tick_exception	HALT.json active:true between ticks	no

MONDAY AFTER CLOSE

rank	action	owner	files	test	rollback	PK decision
16	X1 qty 1→2 with partial-fill handling, qty-aware symmetry/exit, store P&L, Qty column, GOV parity, docs	Claude	governance.json, risk.py, loop.py, store.py, app.py, 4 test files, 3 docs	21 X1 tests + full pytest + <code>--once</code> <code>--dry-run</code>	revert fixed_quantity to 1	yes — precondition (no Monday fill)
17	Phantom-open reconcile via list_orders(closed, —limit 500) + summed filled_qty; no synthetic; 15:45 rung → summary ok=False	Claude	loop.py, alpaca.py, test_loop.py	Thu 14:35 scratch → exit_ filled qty 2 debit 0.31; two-order case → 0.32; none → al- ready_flat only	git revert	no
18	poll_until_filled: assert once + bare order get + monotonic deadline; writes still assert	Claude	alpaca.py, loop.py, test_agent.py	17 spawns per 15-attempt poll (was 48); dead- line honoured at 5 s/attempt	git revert	no
19	Tick-start sweep of open tg-e-* through <code>_cancel_and_confirm</code> (dry_run-aware), raced fill → entry_filled from entry_intent	Claude	loop.py, test_loop.py	10:50 tick cancels s1; filled → man- aged; dry_run → no cancel	git revert	no
20	available_underlyings in brain context + no_trade backstop + skip fetches when empty	Claude	loop.py, brain.py, tests	after SPY fill con- text lists QQQ; SPY pick → un- derlying_unavail- able; both open → no propose	git revert	no
21	Dashboard batch: level-based whitelist + proposal/partial/mcp events, drop outside_entry_window rows, alerts panel, visible HALT reason (escaped), Detail fallbacks, Proposer-vs-Guard section, guarded governance read, weekday sessions, merge PR #9	Claude	store.py, app.py, test_store.py, test_app.py	parity test greps loop.py levels; AppTest shows red panel + rea- son; 80 filler rows do not bury a crit- ical; malformed governance → empty state	git revert	no
22	X2 MCP reconcile script + reconcile.yml + 16 tests + .mcp.json pin + README paragraph; dispatch once after landing	Claude	scripts/mcp_recon- cile.py, reconcile.yml, test_mcp_recon- cile.py, .mcp.json, README.md	16 tests + live dispatch: con- nected, ≤ 6 turns, one event pushed	<code>gh work- flow disable</code>	yes — new deps (uv, al- paca-mcp- server==2.3.0), mismatch exit code
23	Optional entry.window_min- utes 25 with .get default	Claude	governance.json, loop.py, test_loop.py	window boundaries	remove key	no

rank	action	owner	files	test	rollback	PK decision
24	Redraw architecture/sequence/flowchart via diagram-design; PLAN:241, HANDOFF:77; diagram-vs-code test	Claude	docs/diagrams/*.html, PLAN, HANDOFF, test_agent.py	stripped text has no Critic/14/15 min/BEAR CALL/0.45/read-only MCP	git revert	yes — PK picks types
25	16:9 cover (fresh render), PR #11 content review, form texts, social post 02	team	deck/theta-gate.tex, social/*	cover 1920x1080; deck grep clean; post live	n/a	yes — public positioning

TUESDAY

rank	action	owner	files	test	rollback	PK decision
26	First qty-2 window watch; verify partial handling; 16:05 reconcile run; local dashboard check	team	—	entry_* qty 2 or partial fields; reconcile green	HALT between ticks	no
27	Untracked-position reconstruction from the closed tge- order (synthetic entry_- filled with every exit-path key, same-tick management, keep HALT)	Claude	loop.py, test_loop.py	untracked pair + closed order → managed this tick, Thu rung submitted, no duplicate	git revert	no
28	Pre-flip checklist, /security-review, HANDOFF purge, flip public + topics, deploy app.py from main as-is, browser-verify, URL into README	PK	README.md, docs/HANDOFF.md	PUBLIC/MIT/topics; URL renders cold on two devices; 5 tabs no error	re-privatise (history stays exposed); delete app	yes — public positioning, irreversible
29	Write-up skeleton with disclosure lines; deck freeze; video host chosen	team	README.md, deck/theta-gate.tex	every number greps to governance or store.summary	n/a	yes

WEDNESDAY

rank	action	owner	files	test	rollback	PK decision
30	09:50 ET gate_check for live VRP; watch the single 10:30 tick; manual dispatch fallback	team	—	proposal + entry_- filled or a named gate reason	HALT before 10:30	no
31	Video fallback screen take after 10:45; docs truth pass (14 drift items)	Claude	PLAN, HANDOFF, STRATEGY, governance comments	grep list returns zero stale hits	n/a	no

THURSDAY

rank	action	owner	files	test	rollback	PK decision
32	Flatten watch 14:30–16:15; confirm exit_filled per rung; 16:05 reconcile; manual close only if the ladder failed (disclosed)	team	—	position list [] by 16:10; every position has close_debit; reconcile clean	manual close per runbook	yes — human order on the judged account
33	Media block: deck rebuild with store.summary numbers, 3.5–4.5 min video, final write-up, cover, upload	team	deck/theta-gate.-tex, README.md	zero PLACEHOLDER; 3:30–4:30; numbers match main	Wednesday take + Thursday voice-over	yes

FRIDAY

rank	action	owner	files	test	rollback	PK decision
34	Submit the form 09:00–10:30 ET; after 16:00 ET disable agent.yml and reconcile.yml	PK	agent.yml, reconcile.yml	confirmation before 11:00; both workflows disabled; no bot commit after 20:05 UTC	gh work-flow enable	yes — irreversible

Designs**X1 — FIXED_QUANTITY 1→2 WITH PARTIAL-FILL HANDLING (LANDS MON AFTER 16:00 ET, FIRST EFFECT TUE 10:30 ET)**

Change list (verified against the code and the throwaway account's order JSON: top-level `qty` / `filled_qty` are strings, canceled orders keep `filled_qty`):

file	function / line	change
governance.json	strategy.fixed_quantity:20; _fixed_quantity_comment:19; risk_max_concurrent_comment:52	1 → 2; comments rewritten (rendered verbatim by app.py:513)
risk.py	size_position:230-244; gate_max_loss_per_trade:249	docstrings/messages only; arithmetic unchanged; note the gate binds only from qty 3
loop.py	new <code>_filled_qty(order, fallback)</code> next to <code>_extract_actual_price</code>	<code>int(float(filled_qty or 0))</code> with try/except
loop.py	<code>_journal_entry_fill</code> :660-671; <code>_journal_exit_fill</code> :430-444	journal filled qty + requested_qty; max_loss from filled qty; close_debit stays filled_avg_price
loop.py	<code>_attempt_entry</code> s0 cancel:700-712 and s1 cancel:735-743	<code>filled_qty > 0</code> → <code>entry_filled(partial=True)</code> , ladder stops, no top-up, no unwind
loop.py	<code>_check_leg_symmetry</code> :332-350	compare <code>abs(qty)</code> per leg, not symbol sets
loop.py	<code>_evaluate_and_exit_position</code> :630-632	<code>_broker_paired_qty</code> ; unequal legs → HALT + <code>exit_reconciliation_gap</code> ; <code>paired ≠ journal qty</code> → <code>position_qty_mismatch</code> warning
loop.py	<code>_attempt_exit</code> s0 cancel:499-504; s1 body/cancel:506-529	<code>partial</code> → <code>exit_partial_fill</code> (informational) and <code>qty</code> → <code>filled</code> before s1; next tick recomputes from broker
loop.py	<code>_attempt_force_close</code> stale-cancel:573-586	journal the partial; refetch positions; recount before the rung body
loop.py	<code>_open_positions</code> :114-123	unchanged: closes on <code>exit_filled</code> only (pinned by a test)
alpaca.py / spread.py	<code>submit_mleg</code> :186; <code>poll_until_filled</code> :242; mleg bodies	unchanged; document that <code>partially_filled</code> is deliberately non-terminal
store.py	<code>_build_positions</code> :280-298; <code>decision_log</code> :449-453	<code>sum partial + final exit debits by qty</code> ; <code>pnl = credit×100×entry_qty − ∑ debit×100×qty</code> ; whitelist <code>exit_partial_fill</code>
app.py	positions table:454-465	Qty column
test_agent.py	GOV:19-68, :229, :424	load governance.json; assert qty 2
test_store.py / test_app.py	fixtures	qty parameter; mixed qty-1/qty-2 fixture
docs	HANDOFF:110; PLAN:180/189/314/328; canonical plan §2.12 amendment	1-then-2 schedule; remove the phantom partial-fill unwind

Arithmetic at width 5, qty 2, credit $c \in 0.50\text{--}1.25$: max loss = $1000 - 200c \leq \$1,000$ for every c (gate never vetoes at qty 2; qty 3 vetoed for $c < 1.667$); margin \$1,000 per position, \$2,000 at two; total open risk < \$3,000 always; BP floor binds only below \$26,000; take-profit +\$60, stop −\$120, full loss −\$880 at $c = 0.60$; daily −1% needs ~8 simultaneous stops; week ceiling $\approx +\$240\text{--}260$ gross.

New failure modes: (1) Tue 10:30 partial s0 + full s1 = 3 lots, journal 2, no HALT (fixed by the s0 branch); (2) partial s0 + unfilled s1 = 1 untracked lot → HALT, never managed (fixed); (3) exit partial → s1 over-close, broker response unverified (fixed by qty reduction); (4) 15:00 rung recount vs step-4 snapshot (fixed by refetch); (5) naked-leg detection loses coverage at $\text{qty} \geq 2$ (fixed by `lqtyl` compare); (6) store P&L ignores exit qty (fixed); (7) all five money gates unreachable by arithmetic

— say so in the write-up; (8) partials become the base case (~57% over 8 orders); (9) mixed-qty journal from Tuesday — never read gov fixed_quantity for an open position; (10) poll budget unchanged, only hit more often; (11) no CI pytest before Tue 09:30 (rank 12 fixes).

Tests: the 21 listed in rank 16 plus the manual pre-push gate. Open questions: canceled-partial status shape; over-close rejection vs acceptance; whether paper mleg partially fills at all; landing without a Monday qty-1 fill; warn vs HALT on journal/broker qty mismatch.

X2 — SCRIPTS/MCP_RECONCILE.PY, READ-ONLY MCP RECONCILIATION (LANDS MON AFTER CLOSE, CRON TUE 16:05 ET)

Prototype verified live against the submission account: server connected, init toolset 5 read + 5 docs tools and zero write tools (with tools=[] alone all ~70 mcp_alpaca__ tools stayed in the toolset — ALPACA_TOOLSETS=trading plus disallowed_tools is what strips the writes), num_turns 3, \$0.04, 5.4 s, one mcp_reconciliation event, HALT.json absent.

file	change
scripts/mcp_reconcile.py	ClaudeAgentOptions(model claude-opus-5, max_turns 6, tools=[], allowed_tools=[get_all_positions, get_orders], disallowed_tools=[10 write tools], permission_mode dontAsk, mcp_servers={'alpaca': uvx alpaca-mcp-server==2.3.0, ALPACA_PAPER_TRADE true, ALPACA_TOOLSETS trading}, strict_mcp_config, setting_sources=[]); data parsed from ToolResultBlocks only (server envelope data.result), never the model's text; failure = status ≠ connected, any tool is_error (a 401 arrives as is_error with the server still connected), None data, result is_error; diff reuses loop._open_positions: matched / journal_only / broker_only / broker_equity / orders_seen / fills_unknown_to_journal / journal_fills_unknown_to_broker; one journal event, level info or warning, rc 0 clean / 1 mismatch or failure; loop._git_publish; tick lock respected (rc 2); never imports alpaca.py, never HALTs
.github/workflows/reconcile.yml	cron '5 20 * * 1-5' + dispatch; same concurrency group as the tick; timeout 8; setup-python 3.14; pip requirements + uv==0.10.4; uvx alpaca-mcp-server==2.3.0 --help warm-up; git identity as agent; yml:76-78; secrets ALPACA_API_KEY/SECRET_KEY/ANTHROPIC_API_KEY; no Alpaca CLI install
store.py:449-453	whitelist mcp_reconciliation, mcp_reconciliation_failed (strict xfail proves the event is invisible otherwise)
test_mcp_reconcile.py	16 tests (options never allow a write tool; refuse live flag; envelope unwrap; results not text; 401 flagged; not connected; diff incl. F3 phantom → journal_only; clean run one event + one publish; mismatch warning rc 1 no HALT; timeout / CLI missing / publish failure)
.mcp.json, README.md	pin ==2.3.0; "MCP is the auditor, not the trader" paragraph

New failure modes: cold uvx resolve slower than the MCP connect window (mitigated by the warm-up); Thu 16:05 red by design when F3 phantoms exist unfixed; 16:05–16:15 journal push collides with human pushes; a dispatch during the session shares the single pending slot; a rotated key surfaces only at 16:05; Anthropic-side failures → failed event, red run; max_turns/150 s caps → failed event; broker cids and OCC symbols land in the public journal (feature); no end date on the cron; quantities are not compared (a -2/+1 partial state reads as matched).

Tests: the 16 above, the promoted test_store case, the live dispatch on the runner. Cost: ≤ \$0.40 per run (first run pays the prompt-cache write), < \$3 for the week.

Audit coverage and gaps

Dimensions not run (from the critic): the broker WRITE contract on the throwaway profile (submit success/--dry-run stdout, 422 duplicate-id, cancel-after-fill 422); the open option position shape (never observed — both accounts hold 0 positions; only position list --schema backs asset_class == "us_option"); Cboe CSVs from a GitHub egress IP; brain.purpose on the actual Linux runner with the repo's key; Anthropic key health; the Monday merge-freeze plan; post-change re-

gression for the two fixed changes; the human remediation runbook; the first-ever scheduled fire; alpaca-mcp-server (until X2's live run); whether the Saturday journal lines stay; whole-tick worst-case wall clock on the runner. Ranks 6, 11, 13, 14, 16, 22 close most of these before their bite dates.

Unchecked claims: entry-path-1, tests-coverage-1, the three F1 records, strategy-pnl-1, entry-path-2, the three F3 records, both F4 highs, F18, entry-path-3, durability-runner-2, broker-contract-3, exit-path-2, broker-contract-2, schedule-time-1/-2, strategy-pnl-2, F5, F19, exit-path-3/-4, F8, F10, F20, F7 were confirmed by verifiers who read code only; the finders (not the verifiers) executed the live calls and scratch tests quoted above, and the F2 verifier ran the live client-id call. hackathon-compliance-1/-2/-6 were re-verified by the critic (PRIVATE, licenseInfo null, no LICENSE, 0 deployments).

Bad citations corrected in this report: entry-path-2 → governance.json:77-83 (not 83-88); hackathon-compliance-1 → `gh repo view` (README:1 is `# Theta Gate`); hackathon-compliance-2 → `ls LICENSE*` (SKILL.md:1 is frontmatter); hackathon-compliance-4 → risk.py:1 carries no compliance claim; docs-vs-code-11 → PLAN.md:251 is a heading about skill filenames, not a client_order_id format; F22 → requirements.txt has 14 lines; tests-coverage-5 → market.py ends at 281; tests-coverage-6 → brain.py ends at 265; F10 → app.py:358 is inside load(), the governance render starts at 509; F4 → `_wait_for_terminal` is loop.py:301-307; schedule-time-1 → step 8 is loop.py:1023-1026; hackathon-compliance-7 → HANDOFF.md:22 covers only the write-up; the three diagram findings → line 1 is `<!DOCTYPE html>`, the evidence is in the bodies. Three verifier records (F3, schedule-time-1, broker-contract-2) cite line ranges from another revision and F3's repeats the false "returns None on miss" premise — the claims stand on other evidence; do not reuse those citations.

Uncited tracked files: .claude/settings.json (allow/deny lists, safe to publish); four .claude/skills symlinks; .gitignore (does not ignore data/loop.lock — harmless, the bot adds named paths); .streamlit/config.toml (showErrorDetails none, stale 'repo is private' comment); data/HALT.json ({“active”: false} — a PR carrying a stale active HALT would block Monday); data/events_2026-09-*.json (five events, tz -04:00, none intersects a window); data/journal.jsonl (13 Saturday lines incl. one critical not_paper_abort); docs/THETA_GATE_CANONICAL_PLAN.md (grepped only); env.example (placeholders; may trip a secret scanner after the flip); skills-lock.json; social/drafts/01; test_app.py (27 of the 96 tests).

Open risks

1. Zero trades Mon/Tue is the base case even after every fix; the week's P&L rides on Wed 10:30–10:45 ET unless PK relaxes the VRP floor (rank 7).
2. The broker write contract (submit/--dry-run stdout, 422 channels, position shape) is inferred from source and help text; rank 14 is the only pre-10:30 retirement.
3. Mon 31 Aug is the first-ever runner execution of the schedule trigger, brain.propose with the runner's key, the Cboe fetch from a GitHub IP and the bundled claude CLI; failures there are green-run silent no_trades unless rank 6 runs tonight.
4. On the runner the CLI ignores the 'submission' profile (ALPACA_API_KEY override); paper rests on ALPACA_LIVE_TRADE=false + assert_paper (verified to pass).
5. qty 2 opens partial fills for the first time (~57% chance of at least one this week); the canceled-partial shape and the over-close response are unobservable until they happen.
6. All five money gates are unreachable by arithmetic at qty 2 / width 5; a post-submit doubled position (\$1,323 max loss) has no gate re-check.
7. Timeout 25 + one pending slot: a long tick cancels the next scheduled tick (grey), gapping exit evaluation to ~10 min.
8. Submission account eligibility (created 3h06m pre-kickoff, \$100k shown a day before created_at) and pre-kickoff code disclosure are PK's calls (rank 13, rank 29).
9. The public flip permanently exposes the 26 Aug note, the Gmail author identity and Actions logs; the ecc-tools App keeps push to unprotected main until PK decides.
10. Three of four mandatory media deliverables do not exist; the deck needs a Thu ~16:15 ET rebuild; form limits (255 chars / 100 words / 300 MB / 5 min) are unverified; the LinkedIn post could not be verified.

11. Rank 22's reconcile job exits 1 on any mismatch — Thu 16:05 ET goes red whenever journal and broker disagree after the flatten (alarm, not proof).
12. Neither cron has an end date; rank 34 disables both after Fri 16:00 ET.
13. data/journal.jsonl carries a Saturday critical and a weekend "session"; kept on purpose pending rank 13(d).
14. Nothing changes the arithmetic edge: credit/width ≈ 0.12 , margin = full width, EV of a fairly priced spread = minus the bid-ask; realistic week $-\$600\dots+\250 .

Method

Agents: 11 finders (entry-path, exit-path, broker-contract, schedule-time, durability-runner, read-model-dashboard, tests-coverage, docs-vs-code, security-public-repo, hackathon-compliance, strategy-pnl); verifiers producing 48 confirmed records (34 at 3/3, 4 at 2/3, 10 at 1/1 — mediums beyond the per-dimension cap of 12 and lows were passed through unverified: 77 records) with up to three fix objections each; 6 refutations; 4 judges (one per criterion, each scoring Theta Gate and 12 lablab competitors from blurbs); 2 designs (X1 sizing, X2 MCP reconciliation — X2 prototyped and run live read-only); 1 critic; 1 synthesizer (this report).

Executed, not read: read-only Alpaca CLI on the submission profile (account, positions, orders open/closed/all, clock, doctor, quotes, bars, option chains at limits 100/1000/10000, page-token round trip, strike bounds, client-id and order-id misses with stdout/stderr/rc split); the throwaway 'paper' profile's four historical orders (mleg shape, filled_avg_price sign); the real Python functions `alpaca.get_order_by_client_id/get_order` (RuntimeError observed) and the production entry pipeline against Friday's data at both chain limits; one to three billed `brain.propose` calls locally (7.6–9.5 s, valid JSON); Cboe CSVs with `urllib`'s UA; CLI v0.0.13 Go source (`root.go`, `client.go`, `factory.go`, `order.go`, `commands.gen.go`); gh API for repo, PRs, runs (3 dispatch runs, 0 scheduled), secrets/variables names, collaborators, licenses; lablab rule book, guidelines, ToS and the X post via `firecrawl`; ~90 scratch `pytest` tests across dimensions, all under the scratchpad with `journal/HALT/lock` redirected; `streamlit 1.62.0` source and `AppTest`; a Python 3.12 `venv` for the dashboard; the X2 prototype end to end against the MCP server.

Read only: `loop.py`, `spread.py`, `risk.py`, `market.py`, `brain.py`, `alpaca.py`, `store.py`, `app.py`, `governance.json`, `agent.yml`, all docs, all diagrams (tag-stripped), the deck `.tex` via PR diff.

Capped or forbidden: no broker writes on any account (submit/cancel shapes inferred); no MCP write tools; no repo or git writes; medium findings beyond 12 per dimension unverified; the LinkedIn post, the deck PDF metadata, the `ecc-tools` App's declared permissions, GitHub notification settings, and the Streamlit Cloud dropdown contents were not observable. Today's check: HEAD 92b2f96, `gh pr list` open = #7 #8 #9 #10 #11.